§3.1: Graphical deformations of curvettas yield fillings

Nishtha Tikalal

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1. Set-up

As covered previously in §1 and §2, recall the following set-up:

Let $(X, 0)$ be a rational surface singularity with a reduced fundamental cycle, and let $(\mathcal{C}, w)$ be its associated decorated plane curve germ with smooth, weighted branches $C_1, C_2, \ldots, C_m$.

- **Graphical Coordinates:** Inside a sufficiently small Milnor ball $B \subset \mathbb{C}^2$, we choose complex coordinates (x, y) such that all branches are strictly graphical: $C_i = \{y = f_i(x)\}$.
- **Smooth Configurations:** We consider smooth graphical arrangements $\Gamma = \{\Gamma_1, \Gamma_2, \ldots, \Gamma_m\}$, where each branch deforms into a smooth graphical disk $\Gamma_i = \{y = g_i(x)\}$ intersecting ∂B transversally.
- **Symplectic Strategy:** The goal is to construct a symplectic analogue to the de Jong–van Straten algebraic framework, showing that Stein fillings of the link of $(X, 0)$ can be obtained via these smooth graphical homotopies of the branches of $\mathcal{C}$.

2. Homotopies

Definition 2.1 (Smooth Graphical Homotopy)**.** Let $(\mathcal{C}, w)$ be a decorated plane curve germ with weights $w_i = w(C_i)$ on its smooth graphical branches $C_1, C_2, \ldots, C_m$. A *smooth graphical homotopy* of $(\mathcal{C}, w)$ is a smooth family of branches C_i^t for $t \in [0, 1]$ (where $\mathcal{C} = \bigcup_{i=1}^m C_i^0$), together with a set of distinct marked points p_k $(k = 1, \ldots, n)$ on $\bigcup_{i=1}^m C_i^1$. Inside a Milnor ball B, the following conditions must hold:

1. **Graphical Boundary Condition:** Each branch is given by $C_i^t = \{y = f_i(x, t)\}$, where $f_i(x, t)$ is smooth, and C_i^t intersects ∂B transversally for all t.
2. **Interior Intersections:** All branch intersections remain strictly inside the interior of B throughout the entire homotopy $(t \in [0, 1])$.
3. **Transversality at Final State:** At $t = 1$, all intersections between any two distinct branches C_i^1 and C_j^1 are completely positive and transverse.
4. **Weight/Marked Point Constraints:** At $t = 1$, all actual intersection points on each branch C_i^1 are marked. Additional "free" marked points may be added to the interior of a unique branch C_i^1 such that the total number of marked points on C_i^1 is precisely equal to its weight w_i.

Instead of mapping disks sequentially in a chain, the homotopy tracks the entire configuration of disks deforming simultaneously over time t. At the starting state $(t = 0)$, the setup begins with a highly singular plane curve germ $\mathcal{C}$ where all branches meet at a complicated singular point inside the Milnor ball B. As the deformation progresses $(t \to 1)$, the homotopy acts as a smooth movie showing these disks shifting and sliding; because they remain strictly graphical $(y = f_i(x, t))$, they never tear, fold, or escape through the boundary of the ball. By the end state $(t = 1)$, the singularity has completely broken apart into a perturbed arrangement of disks $\Gamma_i = C_i^1$ that only cross each other cleanly, transversely, and positively. Finally, the weights w_i act as a budget for each branch: at the final state, every physical crossing on a given disk is recorded as a marked point, and if this total count falls short of w_i, extra standalone free points are added to the interior of that disk until the budget is met exactly.

For small homotopies (where t is close to 0), the graphical and interior intersection constraints are automatically satisfied due to geometric closeness and coordinate choice. Because the perturbed branches C_i^t remain C^1-close to the original singular branches C_i, their tangent spaces stay confined within a small neighborhood of the finite tangent directions of $\mathcal{C}$ at the origin. By selecting a projection coordinate system whose fibers completely avoid these few tangent directions in the Grassmannian, the branches are guaranteed to remain strictly graphical and intersect the boundary ∂B transversally (Condition 1). Furthermore, since the Milnor ball B is intentionally chosen to isolate only the intersection of the branches at the origin, a sufficiently small perturbation ensures that the migrating intersection points cannot drift far enough to escape the ball, meaning they must remain trapped within the interior of B (Condition 2). This doesn't hold well for larger homotopies.

3. Picture Deformations

Definition 3.1 (Picture Deformation)**.** A *picture deformation* $\mathcal{C}^S$ of the decorated germ $(\mathcal{C}, w)$ with smooth components $C_1, \ldots, C_m$ over a germ of smooth curve $(S, 0)$ is given by:

1. A δ-constant deformation $\mathcal{C}^S \to S$ of the plane curve germ $\mathcal{C}$, and
2. A flat deformation $w^S \subset \widetilde{\mathcal{C}^S} = \widetilde{\mathcal{C}} \times S$ of the zero-dimensional scheme w,

such that for a generic fiber $s \neq 0$:

- The divisor w^s is completely reduced.
- The only singularities of the deformed curve $\mathcal{C}^s$ are ordinary multiple points.
- The set of ordinary multiple points m^s is entirely contained within the support of the reduced divisor w^s (i.e., $m^s \subset w^s$).

To mirror the algebraic framework where the zero-dimensional scheme w deforms into a reduced divisor w^s for $s \neq 0$, we explicitly track the marked points as functions of the time parameter $t \in [0, 1]$:

- **Initial State ($t = 0$):** The marked points are concentrated entirely at the origin on each branch, providing the initial configuration/decoration of the singular curve germ $(\mathcal{C}, w)$.
- **Dynamic Motion ($0 < t < 1$):** We define smooth path functions $p_i^j(t) \in C_i^t$ for $j \in \{1, \ldots, w_i\}$ that describe the continuous movement of these marked points along the deforming branches during the homotopy.
- **Final State ($t = 1$):** The end-state positions $p_i^j(1) = p_i^j$ satisfy the condition that all actual physical crossings are captured by these points. Consequently, at $t = 1$, the branch C_i^1 contains at most w_i total intersections with other components.

It's pretty clear what is happening at $t = 0$ and $t = 1$. However, for the intermediate constraints, it is not as clear. Mechanistically, this is important for showing the existence of unexpected Stein fillings.

- **Topological Freedom ($0 < t < 1$):** During a smooth graphical homotopy, the tracking paths $p_i^j(t)$ carry no structural or geometric weight prior to the endpoint. The deforming curvettas C_i^t are free to interact dynamically without constraints. They can generate an arbitrary number of temporary intersections, and these crossings are permitted to have either *positive* or *negative* orientations.
- **The Algebraic Constraint:** In contrast, a picture deformation is governed by complex analytic geometry. Because the components are deformed through complex algebraic curves:
  1. Every intersection must be strictly positive (int > 0) due to the natural orientation of complex submanifolds.
  2. The intersections between branches must be explicitly marked at all times.
  3. The total intersection count remains tightly bounded and constant, dictated continuously by the weights w_i.

By shifting from the rigid algebraic category to the flexible smooth category, the intermediate states are allowed to cross algebraic barriers (e.g., creating and canceling mixed-sign intersection pairs). As long as the system clears into a strictly positive, transverse configuration at the $t = 1$ checkpoint, it is sufficient to build a valid Stein filling.

4. Lefschetz Fibrations

Lemma 4.1. *Let $\Gamma_1, \dots, \Gamma_m$ be smooth disks in $\mathbb{C}^2$ which are graphical with respect to the projection π_x, so $\Gamma_i = \{y = f_i(x)\}$. Assume that at each intersection point of two or more Γ_i, there exists a neighborhood U of the intersection such that $\bigcup_i \Gamma_i$ are cut out by complex linear equations inside U. (Up to graphical isotopy, this only requires the Γ_i to intersect transversally and positively with respect to the orientation on the graph Γ_i induced from the natural orientation on $\mathbb{C}$.) Let $p_1, \dots, p_n$ be points on the disks Γ_i which include all intersection points, and let $\alpha : \mathbb{C}^2 \# n\overline{\mathbb{C}P^2} \to \mathbb{C}^2$ be the blow-up at the points $p_1, \dots, p_n$. Let $\widetilde{\Gamma}_1, \dots, \widetilde{\Gamma}_m$ denote the proper transforms of $\Gamma_1, \dots, \Gamma_m$.*

Then $\pi_x \circ \alpha : (\mathbb{C}^2 \# n\overline{\mathbb{C}P^2}) \setminus (\widetilde{\Gamma}_1 \cup \dots \cup \widetilde{\Gamma}_m) \to \mathbb{C}$ is a Lefschetz fibration whose regular fibers are punctured planes, where each puncture corresponds to a component $\widetilde{\Gamma}_i$. There is one vanishing cycle for each point p_j, which is a curve in the fiber enclosing the punctures that correspond to the components Γ_i passing through p_j.

Similarly, if $B = D_x \times D_y$ is a Milnor ball that contains all the points $p_1, \dots, p_n$ and contains $(D_x \times \mathbb{C}) \cap (\bigcup_i \Gamma_i)$, and T_i is a small tubular neighborhood of $\widetilde{\Gamma}_i$, then $\pi_x \circ \alpha : (\alpha^{-1}(D_x \times D_y)) \setminus (T_1 \cup \dots \cup T_m) \to D_x$ is a Lefschetz fibration with compact fiber. The fiber is a disk with holes corresponding to the components Γ_i. The vanishing cycles correspond to the points p_j in the same way.

If the curvettas $C_1^1, \dots, C_m^1$ with marked points are the result of picture deformation of the germ $(\mathcal{C}, w)$ associated to a surface singularity, the Lefschetz fibration constructed as above is compatible with the complex structure on the Milnor fiber of the corresponding smoothing.

To visualize, think of the initial setup as a flat base disk D_x parameterized by x, where above every point $x \in D_x$, the fiber of the projection π_x is a vertical complex plane $\mathbb{C}_y$. The graphical disks Γ_i act as a collection of smooth, non-vertical sheets that float above D_x, occasionally crashing into one another at specific isolated intersection points p_j. When we perform a smooth blow-up α at an intersection point p_j, we are replacing that point with an exceptional sphere $\mathbb{C}P^1$; geometrically, this separates the crashing sheets by separating their different tangent slopes, lifting them into well-behaved, disjoint proper transforms $\widetilde{\Gamma}_i$. Crucially, because we then completely drill out these proper transforms (taking the complement), every sheet $\widetilde{\Gamma}_i$ becomes a missing puncture in the vertical fibers. If we look at a generic vertical fiber over a point x_0, it looks like a disk with m holes (one hole drilled out for each sheet). However, as our base coordinate x moves toward the shadow of a blown-up point p_j, the holes corresponding to the sheets that originally met at p_j begin to pinch together; the vanishing cycle is simply a closed loop in the fiber that lassoes these specific crashing holes, shrinking down to a single point at the critical fiber to form a classical Lefschetz critical point. The proof is briefly detailed below.

The blow-up map $\alpha : \mathbb{C}^2 \# \overline{\mathbb{CP}}^2 \to \mathbb{C}^2$ is the projection onto the first factor, sending $((x, y), [u : v]) \mapsto (x, y)$, subject to the relation $xv = yu$. The node (intersection of the exceptional divisor E and the proper transform $\widetilde{F}$) occurs at $((0, 0), [0 : 1])$. Since the second projective coordinate is non-zero ($v \neq 0$), we work in the affine chart where $v = 1$.

Substitute $v = 1$ into the defining equation of the blow-up:

$$\begin{aligned} xv &= yu \\ x(1) &= yu \\ x &= yu \end{aligned}$$

Because x is completely determined by y and u, the coordinates on this chart of the blow-up are parameterized by $(y, u) \in \mathbb{C}^2$. The blow-up map α in these local coordinates is explicitly given by:

$$\alpha(y, u) = (yu, y)$$

The original projection map is $\pi_x(x, y) = x$. Composing this with our local parameterization $\alpha(y, u)$

yields:

$$\begin{aligned}(\pi_x \circ \alpha)(y,u) &= \pi_x(\alpha(y,u)) \\ &= \pi_x(yu, y) \\ &= yu\end{aligned}$$

This confirms that the local expression for the fibration near the node is exactly the standard complex Morse/Lefschetz singularity $f(y,u) = yu$. Let $\Gamma_i = \{(x, f_i(x))\}$ be the smooth sections in $\mathbb{C}^2$ centered at $p_j = (0,0)$. To find their lifts to the blow-up, we substitute $y = f_i(x)$ into the projective coordinate relation $[u : v] = [x : y]$.

For points away from the origin, x and y cannot both be zero.

1. Substitute $y = f_i(x)$ directly into the projective relation:

$$[u : v] = [x : f_i(x)]$$

2. Since $f_i(0) \neq 0$, the ratio is always well-defined, even at $x = 0$ where $[u : v] = [0 : f_i(0)] = [0 : 1]$.

3. Therefore, the curve lifts homeomorphically (isomorphically) to the blow-up as:

$$\widetilde{\Gamma}_i = \{((x, f_i(x)), [x : f_i(x)])\}$$

When $f_i(0) = 0$, both x and y vanish at the origin, meaning the direct substitution $[0 : 0]$ is undefined on the exceptional divisor E. We find the proper transform $\widetilde{\Gamma}_i$ by taking the closure of the lifted curve away from $x = 0$.

1. For any $x \neq 0$, the projective coordinate is given by:

$$[u : v] = [x : f_i(x)]$$

2. To see where this curve intersects the exceptional divisor E (where $x = 0$), scale the projective coordinates by $1/x$:

$$[u : v] = \left[1 : \frac{f_i(x)}{x}\right]$$

3. Take the limit as $x \to 0$ (using a dummy variable $a \to x$ for a moving point along the section) to determine the unique point of intersection on E:

$$[u : v]\Big|_{x=0} = \left[1 : \lim_{a \to x} \frac{f_i(a)}{a}\right]$$

4. Thus, the total transform is the union of the exceptional divisor E and the proper transform:

$$E \cup \widetilde{\Gamma}_i = \{((0,0), [u : v])\} \cup \left\{\left((x, f_i(x)), \left[1 : \lim_{a \to x} \frac{f_i(a)}{a}\right]\right)\right\}$$

We now verify why these proper transforms become disjoint:

1. **Avoidance of the Node:** The node is located at $((0,0), [0 : 1])$. For any curve Γ_i passing through the origin, its proper transform hits E at a point where the u-coordinate is 1 (i.e., $[1 : m]$). Since $1 \neq 0$, these proper transforms never pass through $[0 : 1]$.

2. **Disjointness on E:** The value $m_i = \lim_{a \to 0} \frac{f_i(a)}{a} = f_i'(0)$ represents the tangent slope of the curve Γ_i at the origin.

3. Because the original sections Γ_i intersect **transversely** at p_j, they must have distinct initial slopes:

$$f_i'(0) \neq f_j'(0) \quad \text{for } i \neq j$$

4. Consequently, each proper transform $\widetilde{\Gamma}_i$ intersects the exceptional divisor E at a distinct projective point $[1 : f_i'(0)]$. They no longer intersect each other, resolving the singularity into disjoint sections.

This completes the construction of the compact Lefschetz fibration by removing small, disjoint tubular neighborhoods T_i around the proper transforms of the sections. Topologically, deleting these neighborhoods converts the punctures in the fiber into physical boundary components (holes). Crucially, this surgical operation preserves both the underlying fibration structure and the vanishing cycles of the singularities. Because the total space is a compact 4-manifold fibering over a disk D_x, the restriction of this structure to the 3-manifold boundary naturally induces a planar open book decomposition.

In the geometric setting of a picture deformation of a surface singularity germ (C, w), the deformed curvettas $C_1^s, C_2^s, \ldots, C_m^s$ precisely satisfy the required transverse intersection hypotheses. Consequently, the Stein structure induced on the total space by this Lefschetz fibration is automatically compatible with the classical complex structure of the Milnor fiber. This compatibility holds fundamentally because the composed projection map $\pi_x \circ \alpha$ is a holomorphic function throughout the construction.

5. Monodromy

This section establishes the connection between a geometric arrangement of disks in a Milnor ball and the algebraic data of a braid/mapping class group element on the boundary.

Definition 5.1. An m-braid is a collection of m disjoint, smoothly embedded curves (strands) in a solid cylinder or a solid torus that progress monotonically with respect to a longitudinal or angular coordinate. On the boundary of the Milnor ball $\partial B \cong S^3$, the boundaries of the graphical disks $\partial\Gamma = \partial\Gamma_1 \cup \cdots \cup \partial\Gamma_m$ form a closed m-braid.

Definition 5.2. A **knot** is a smooth embedding of a single circle S^1 into a 3-manifold (here, S^3). An **unknot** (or trivial knot) is a knot that is ambient isotopic to a standard geometric circle in $\mathbb{R}^3 \subset S^3$. In this setup, each individual boundary component $\partial\Gamma_i$ is an unknot, though the collective components are non-trivially linked.

Definition 5.3. The braid monodromy ϕ_Γ is the element of the braid group on m strands (or equivalently, the mapping class group $MCG(\mathbb{C}_m)$ of the m-punctured plane) obtained by tracking the movement of the m intersection points of the strands in the fiber as one traverses the boundary circle $\partial D_x \subset \mathbb{C}$.

The core purpose of this setup is to translate a complex 4-dimensional geometric object (the graphical arrangement Γ inside the 4-ball B) into purely topological and algebraic boundary data (the braid $\partial\Gamma$ and its monodromy ϕ_Γ in S^3).

By restricting the standard coordinates to the boundary circle ∂D_x, the projection $\pi_x : \mathbb{C}^2 \setminus \bigcup_{i=1}^m \Gamma_i \to \mathbb{C}$ restricts to a fiber bundle over S^1 whose fiber is the m-punctured plane $\mathbb{C}_m$. The monodromy of this bundle is precisely the braid monodromy ϕ_Γ.

When constructing the Lefschetz fibration, the required blow-ups are performed strictly at points p_i projecting to the *interior* of the disk D_x. Because these modifications happen deep inside the 4-ball, they leave the boundary behavior over ∂D_x completely untouched.

Consequently, the non-compact version of the Lefschetz fibration inherits the exact same boundary monodromy ϕ_Γ.

To transition from the non-compact version of the space to the compact version, we trim the boundaries to keep the geometric objects finite and manageable. In the non-compact setting, the general fiber is an infinite complex plane with m punctures, denoted by $\mathbb{C}_m$. To make this compact, we replace the infinite plane with a bounded geometric disk D_y, and we replace the sharp punctures with actual physical holes. This leaves us with a compact fiber called P_m, which is topologically a disk with m holes (often visualized as a "Swiss cheese" disk).

The boundary of this compact 4-dimensional total space, denoted by $\mathcal{L}^s$, naturally divides into two geometric pieces that fit together to form a 3-dimensional open book decomposition. The first piece is the horizontal boundary $\partial P_m \times D$, which acts as the **binding** of the open book—the static spine that holds all the pages together. The second piece is the vertical boundary, which is a fiber bundle over the circle $S^1 = \partial D_x$. The fibers here are the pages P_m, and as you travel around the base circle, the pages twist and

glue back onto themselves. This swirling structure forms the **mapping torus**, representing the turning pages of our open book.

The core insight is how the twisting (the monodromy) of the compact pages relates back to the twisting of the non-compact punctures. Because the geometric blow-up operations performed earlier happen strictly deep inside the interior of the space, the blow-up map α behaves completely like the identity map when evaluated out at the boundary circle ∂D_x. This means the boundary bundle does not actually care whether we are looking at the infinite punctured plane or the compact disk with holes.

The commutative diagram visualizes this perfect geometric alignment:

$$\begin{array}{ccc} P_m & \xrightarrow{\phi} & P_m \\ \downarrow & & \downarrow \\ \mathbb{C}_m & \xrightarrow{\phi_\Gamma} & \mathbb{C}_m \end{array} \tag{1}$$

The top horizontal arrow, ϕ, represents how the compact page P_m wraps around the boundary loop, while the bottom horizontal arrow, ϕ_Γ, represents how the non-compact punctured plane $\mathbb{C}_m$ wraps around. The vertical arrows are simple inclusions, showing that P_m sits naturally inside $\mathbb{C}_m$. Because the diagram commutes, it proves that the monodromy of our newly constructed open book ϕ is completely identical to the braid monodromy ϕ_Γ we started with. This gives us a massive advantage: we can now use algebraic tools from braid groups and mapping class groups to directly study the 3-dimensional contact topology on the boundary.

Lemma 5.4. *Let $\Gamma = \{\Gamma_1, \dots, \Gamma_m\}$ be a smooth graphical arrangement with marked points $\{p_j\}$, and $B = D_x \times D_y$ a Milnor ball whose interior contains all marked points, such that $\Gamma_i \cap (D_x \times \mathbb{C}) \subset B$ and Γ_i is transverse to ∂B for all $i = 1, \dots, m$. Let ϕ_Γ be the monodromy of the braid $\partial\Gamma = \partial\Gamma_1 \cup \cdots \cup \partial\Gamma_m \subset \partial B = S^3$.*

Let $\phi : P_m \to P_m$ be the monodromy of the open book induced by the Lefschetz fibration constructed for $(\Gamma, \{p_j\})$ in Lemma 3.2. Then ϕ_Γ is the image of ϕ under the projection

$$\eta : MCG(P_m) \to MCG(\mathbb{C}_m)$$

induced by the inclusion $P_m \hookrightarrow \mathbb{C}_m$ of the compact disk with m holes into the m-punctured plane.

When the constructed geometric arrangement $(\Gamma, \{p_j\})$ can be continuously and smoothly deformed into the classical decorated germ $(\mathcal{C}, w)$ via a graphical homotopy, the boundary behavior remains topologically stable. Specifically, this continuous deformation between the interior disks induces a smooth isotopy between their respective boundary braids, $\partial\Gamma$ and $\partial\mathcal{C} = \partial C_1 \cup \cdots \cup \partial C_m$, on the boundary of the Milnor ball S^3. Since isotopic braids share identical algebraic properties, the braid monodromy ϕ_Γ of our arrangement is exactly equal to the braid monodromy of $\partial\mathcal{C}$. By definition, this boundary data coincides with the classical monodromy φ of the singular point of $\mathcal{C}$, providing an explicit bridge between the deformed Lefschetz fibration structure and the original algebraic curve singularity.To understand this concept intuitively, it helps to view a "smooth graphical homotopy" as a slow-motion movie that continuously deforms one geometric object into another. In this setup, we are taking our specially arranged interior disks Γ and smoothly shifting them until they line up perfectly with the classical algebraic curve germ $\mathcal{C}$ inside the Milnor ball.

We can visualize the boundary $\partial B \cong S^3$ as a solid transparent sphere where the ends of these interior disks form strands (braids) on the surface. As the disks deform deep within the interior, their boundary strands on the sphere simply slide around smoothly without ever snapping, breaking, or crossing through one another. In topology, this smooth, non-disruptive sliding of knots or braids is called an **isotopy**.

The idea is that if two braids can be smoothly deformed into one another without cutting the strands, they are topologically the exact same braid. Because they are the same braid, they must share the exact same wrapping behavior, which means the braid monodromy ϕ_Γ of our manufactured arrangement is completely identical to the classical braid monodromy of $\partial\mathcal{C}$.

Lemma 5.5. *Let $\{\Gamma_1, \dots, \Gamma_m\}$ and $\{\Gamma'_1, \dots, \Gamma'_m\}$ be two smooth graphical arrangements, such that the boundary braid of are braid-isotopic (respecting labels) and the weights on the corresponding components agree. Let $\mathcal{L}$ and $\mathcal{L}'$ be the corresponding Lefschetz fibrations constructed in Lemma 3.2. Then the induced open book decompositions on the boundary have the same page and same monodromy.*

It helps to think of a surface singularity $(X, 0)$ as a four-dimensional space with a sharp corner at the origin. If we intersect this space with a small sphere around the origin, we get a three-dimensional boundary manifold Y, called the *link* of the singularity, equipped with a geometric structure ξ called a contact structure. A central question in low-dimensional topology is finding all the smooth four-dimensional spaces (called *Stein fillings*) that can have (Y, ξ) as their boundary. Through the work of de Jong and van Straten, the authors translate this four-dimensional problem into a two-dimensional picture: studying a collection of intersecting curves in the complex plane, denoted by a weighted plane curve germ $(\mathcal{C}, w)$.

Definition 5.6. A *picture deformation* is a rigid, complex-algebraic way of wiggling a singular plane curve germ $(\mathcal{C}, w)$ so that its complicated intersection at the origin splits apart into simpler, transverse intersection points. Because this deformation comes from complex algebra, it directly corresponds to smoothing out the four-dimensional surface singularity $(X, 0)$ into a smooth space called a *Milnor fiber*.

Definition 5.7. A *smooth graphical homotopy* is a much more flexible, purely topological way of deforming the branches of $(\mathcal{C}, w)$ into a new arrangement (Γ, w). Throughout this continuous motion, the curves are allowed to move without requiring complex-algebraic equations, provided that the weights w on the components stay fixed and the curves do not change how they braid around the boundary.

What is shown is that starting from any arrangement (Γ, w) obtained via a smooth graphical homotopy, one can construct a four-dimensional space equipped with a nice geometric projection called a *Lefschetz fibration* $\mathcal{L}'$. However, knowing that $\mathcal{L}'$ exists is not quite enough; we also need to prove that its three-dimensional boundary matches our original singularity link (Y, ξ) and not some completely different three-manifold.

Definition 5.8. An *open book decomposition* is a way of describing a three-dimensional manifold by imagining it as a collection of two-dimensional surfaces (called *pages*) fanned out around a central curve or link (called the *binding*), much like the pages of a book attached to its spine. Every Lefschetz fibration naturally induces an open book decomposition on its boundary, which in turn determines the contact structure on that boundary.

This lemma shows that the open book decomposition induced on the boundary of $\mathcal{L}'$ depends on the boundary braid of the curve arrangement and the weights w assigned to each branch. Because a smooth graphical homotopy keeps both the boundary braid and the weights completely constant from start to finish, every smooth graphical homotopy of $(\mathcal{C}, w)$ produces a Lefschetz fibration with the exact same boundary open book decomposition.

Every rational surface singularity is guaranteed to have at least one genuine picture deformation. For that specific picture deformation, $\mathcal{L}'$ is the compactified Milnor fiber of the singularity, whose boundary is known to be the link (Y, ξ). Since *any* smooth graphical homotopy induces the exact same boundary open book as this benchmark picture deformation, every smooth graphical homotopy must also fill (Y, ξ).

The proof compares two Lefschetz fibrations: $\mathcal{L}$, which comes from the original singular curve germ $\mathcal{C}$, and $\mathcal{L}'$, which comes from the deformed curve arrangement Γ. Both fibrations induce open book decompositions on their three-dimensional boundaries whose pages are planar surfaces P_m, meaning a two-dimensional disk with m holes punched out (one hole for each curve branch $\Gamma_1, \ldots, \Gamma_m$). The entire goal is to show that the monodromy maps $\phi, \phi' \in MCG(P_m)$ of these two open books are identical.

Definition 5.9. For a surface S, the *mapping class group* $MCG(S)$ is the group of all orientation-preserving symmetries (diffeomorphisms) of S that fix the boundary pointwise, considered up to continuous deformation (isotopy).

Definition 5.10. A *Dehn twist* along a closed loop γ in S is a fundamental symmetry formed by cutting S open along γ, twisting one side by a full 360° turn, and gluing it back together.

In the first step, we relate surface symmetries to braids using a map. If we cap off each of the m holes of P_m by gluing in a disk containing a single marked point, the surface P_m becomes a standard disk with m marked points, denoted $\mathbb{C}_m$. This geometric operation induces a surjective group homomorphism $\eta : MCG(P_m) \to MCG(\mathbb{C}_m)$ from the mapping class group of the planar surface to the braid group on m strands. By Lemma 3.3, both open book monodromies ϕ and ϕ' project to the exact same braid monodromy $\varphi \in MCG(\mathbb{C}_m)$, which gives the identity

$$\eta(\phi) = \varphi = \eta(\phi'). \tag{2}$$

In the second step, we measure the exact difference between ϕ and ϕ' using the kernel of η. Because η is a group homomorphism and $\eta(\phi) = \eta(\phi')$, the composition $\phi^{-1} \circ \phi'$ must lie in $\ker(\eta)$. When we fill in a hole with a marked point, the only geometric information lost is twisting around the boundary of that hole: spinning a single marked point in $\mathbb{C}_m$ does nothing to a braid, whereas twisting parallel to a boundary component in P_m is a non-trivial diffeomorphism relative to the boundary. Therefore, $\ker(\eta)$ is generated by the boundary-parallel Dehn twists $T_1, T_2, \ldots, T_m$ around each of the m holes. Moreover, because these boundary twists happen inside tiny isolated collars around each hole, they commute with every element in $MCG(P_m)$ and lie in the center of the group. This yields the factorization

$$\phi' = \phi \circ T_1^{\alpha_1} \circ T_2^{\alpha_2} \circ \cdots \circ T_m^{\alpha_m}, \tag{3}$$

for some integers $\alpha_1, \ldots, \alpha_m \in \mathbb{Z}$. Proving $\phi' = \phi$ now reduces to proving that $\alpha_i = 0$ for every $i = 1, \ldots, m$.

Definition 5.11. A *lantern relation* is a classic algebraic identity in a mapping class group stating that a product of four positive Dehn twists around the four boundary curves of a sphere with four holes equals a product of three positive Dehn twists around specific interior curves. On planar surfaces, every possible algebraic relation among positive Dehn twists is generated by lantern relations.

In the third step, we count enclosing Dehn twists to isolate each exponent α_i. A Lefschetz fibration encodes its monodromy $\psi \in MCG(P_m)$ as a factorization Ψ into a product of positive Dehn twists along vanishing cycles. For a fixed hole h_i (where $i \in \{1, \ldots, m\}$), define $n_i(\Psi)$ to be the total number of Dehn twist curves in Ψ that enclose the hole h_i, counting both boundary-parallel twists and larger interior twists. By a theorem of Margalit and McCammond, all relations among positive Dehn twists in $MCG(P_m)$ are generated by lantern relations, and any hole lying inside a lantern is enclosed by the exact same number of curves on both sides of the relation. Consequently, $n_i(\psi)$ is a well-defined invariant of the mapping class ψ itself, completely independent of the choice of factorization. Since T_i is a single positive Dehn twist enclosing only the i-th hole and no others, composing ϕ with $T_1^{\alpha_1} \cdots T_m^{\alpha_m}$ adds α_i twists around hole i, giving

$$n_i(\phi') = n_i(\phi) + \alpha_i. \tag{4}$$

In the final step, we match these twist counts with the branch weights w_i to conclude that $\alpha_i = 0$. In the construction of the Lefschetz fibrations $\mathcal{L}$ and $\mathcal{L}'$, vanishing cycles arise from two sources: intersection points between different branches, and blow-ups at free marked points p_i sitting on each branch away from the intersections. Each blow-up at a free marked point on the i-th branch produces a boundary-parallel Dehn twist enclosing the i-th hole, and by construction, the total number of vanishing cycles enclosing the i-th hole equals the weight w_i assigned to the i-th component C_i of the original curve germ $\mathcal{C}$. Because a smooth graphical homotopy keeps the weights w_i constant throughout the deformation, both monodromies have the exact same enclosing twist count:

$$n_i(\phi) = w_i = n_i(\phi'). \tag{5}$$

Substituting this equality into our previous equation yields $w_i = w_i + \alpha_i$, which forces $\alpha_i = 0$ for all $i = 1, \ldots, m$. Thus $\phi' = \phi$, proving that the boundary open book decompositions of $\mathcal{L}$ and $\mathcal{L}'$ are identical.

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